

LYMPHOID NEOPLASIA

Mezigdomide reverses T-cell exhaustion through degradation of IKZF1/IKZF3 and reinvigoration of cytokine production pathways

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KEY POINTS

- IKZF1 and IKZF3 promote T-cell exhaustion via epigenetic control, leading to reduced cytokines and increased exhaustion genes.
- Mezigdomide-mediated degradation of IKZF1/IKZF3 restores function in Tex, leading to enhanced B-cell maturation antigen TCE activity.

T-cell exhaustion has been shown to be a key resistance mechanism to efficacy of T-cell engagers (TCE) in multiple myeloma (MM). Mezigdomide, a potent cereblon E3 ligase modulator that targets IKZF1 and IKZF3 simultaneously for proteasomal degradation, has been shown to modulate T-cell activity in patients with MM. We explored the possibility that targeting IKZF1/IKZF3 could address T-cell exhaustion and restore functionality. We conducted extensive transcriptomic and epigenetic profiling on ex vivo generated exhausted T cells (Tex), using their autologous activated T cells as a comparison. Our study revealed that IKZF1 and IKZF3 are critical regulators that contribute to the development and maintenance of T-cell exhaustion. They regulate transcription by directly binding to promoters and enhancers, both proximal and distal, thereby altering transcriptional potential. Increased IKZF1 binding to exhaustion genes after multiple T-cell stimulations results in the enhancement of transcription, whereas binding to cytokine genes results in transcription repression. Mezigdomide treatment in Tex results in decreased expression of exhaustion-related markers, increased proinflammatory cytokine expression, and enhanced target cell killing with alnuctamab, a B-cell maturation antigen targeting TCE. This study provides crucial mechanistic insights into the roles

of IKZF1/IKZF3 in T-cell exhaustion, supporting the rationale for combining mezigdomide with TCE to enhance therapeutic outcomes in MM.

Introduction

Multiple myeloma (MM) is a hematologic malignancy characterized by the clonal proliferation of neoplastic plasma cells in the bone marrow. In recent years, T-cell–redirecting therapies, such as chimeric antigen receptor T cells and T-cell–engaging (TCE) bispecific antibodies, have shown remarkable efficacy.^{1,2} However, the clinical success of these therapies is often limited by T-cell exhaustion, a state of dysfunction characterized by reduced production of effector cytokines and function, which has been shown to correlate with inferior responses, highlighting the need for strategies to overcome this challenge.^{3–6}

Cereblon E3 ligase modulators represent a new class of therapeutic agents that build upon the success of immunomodulatory agents such as lenalidomide and pomalidomide.

Mezigdomide, a novel cereblon E3 ligase modulator, is a rapid and potent degrader of IKZF1 and IKZF3 currently under clinical investigation.^{7,8} IKZF1 and IKZF3 are important transcription factors that are essential for the proliferation and survival of MM cells.⁹ Additionally, their transcriptional activity exerts a key role in T-cell differentiation and activation.^{10,11} The simultaneous degradation of IKZF1 and IKZF3 by mezigdomide has been associated with increased proliferation and activation of effector T cells in the bone marrow in clinical studies.^{12,13}

IKZF1 and IKZF3 regulate gene expression through the formation of homo- and heterodimers^{14–16} and as part of both active and repressive chromatin remodeling complexes, such as the switch/sucrose nonfermentable complex and the nucleosome remodeling and deacetylase complex.^{16–18} This unique property allows for bidirectional gene regulation,

depending on the recruiting complex. However, the role of IKZF1 and IKZF3 in T-cell exhaustion has not been fully elucidated. In this study, we define the mechanisms by which IKZF1 and IKZF3 contribute to exhaustion and investigate whether mezigdomide can overcome this phenotype associated with therapeutic resistance by restoring T-cell functionality in the context of TCE combination therapy.

Materials and methods

T cells

Total T cells were enriched from buffy coats obtained from healthy donors (New York Blood Center) using RosetteSep Human T Cell Enrichment Cocktail (Stemcell Technologies). Viable frozen peripheral blood mononuclear cells (PBMCs) from untreated patients with MM were obtained from Icahn School of Medicine at Mount Sinai and BioIVT.

Ex vivo generation of exhausted T cells (Tex)

Adopted from previous reports,^{19,20} viable frozen T cells were thawed and stimulated with Dynabeads Human T-Activator CD3/CD28 beads for a course of 14 to 17 days, with complete replacement of medium and Dynabeads every 3 to 4 days.

Flow cytometry and cytokine panel analysis to confirm T-cell exhaustion

Surface and intracellular expression of exhaustion markers were analyzed by flow cytometry. See the source table in the supplemental Methods, available on the *Blood* website, for specific antibody clones. Cytokine and chemokine panel analyses were determined from supernatants collected from longitudinal time points during the exhaustion protocol. Luminex multiplex cytokine/chemokine kits (EMD Millipore) were used according to the manufacturer's instructions and data were acquired using Luminex Magpix (EMD Millipore).

Chromatin immunoprecipitation sequencing (ChIP-seq)

Cells were collected, fixed, stored, and shipped to Active Motif according to the manufacturer's instructions. Active Motif prepared chromatin, performed ChIP reactions, generated libraries, and sequenced the libraries.

RNA-seq

Cells for RNA sequencing (RNA-seq) were prepared according to instructions from Discovery Life Sciences. Five million cells per sample were collected, washed, fixed, and stored at -80°C before shipment to Discovery Life Sciences, where downstream RNA-seq was carried out.

ATAC-seq

Cells for assay for transposase-accessible chromatin sequencing (ATAC-seq) were prepared according to instructions from Novogene. Five million cells were collected and cryopreserved before shipment to Novogene. ATAC-seq was performed as previously reported.²¹

Hi-C

High-throughput chromosome conformation capture (Hi-C) was generated by Arima Genomics using the Arima-Hi-C kit

(A510008) according to the manufacturer's protocols, which was adapted from a previous report.²²

Single-cell multiome ATAC + gene expression sequencing

Library preparation for single-cell multiome was conducted following the manufacturer's instructions (10x Genomics, CG000338 Rev F). Quality control of the libraries after construction and complementary DNA quantification were assessed using an Agilent Bioanalyzer High Sensitivity chip (Agilent). Initial quality control of the libraries was performed using Illumina MiSeq paired-end sequencing. Both single-cell RNA-seq (scRNA-seq) and scATAC-seq libraries were sequenced using Illumina NovaSeq 6000 paired-end sequencing. Expression of known marker genes was manually inspected, and cell types were assigned to the initial clusters for the 3 cell atlases of chromatin accessibility and gene expression, independently and jointly.

T-cell cytotoxicity assays

T-cell cytotoxic coculture assays with Cytotox (Sartorius)-labeled, B-cell maturation antigen (BCMA)-expressing K562 target cells in the presence of alnuctamab, a TCE, were performed using Incucyte (Sartorius) over a course of 5 to 6 days.

More detailed experimental procedures can be found in the supplemental Methods.

Results

Ex vivo generation of Tex and demonstration of reduced cytotoxic capacity in a coculture model

To compare the distinct epigenetic states between activated and Tex, established protocols were used to generate an ex vivo model of T-cell exhaustion.^{20,23} In this model, total T cells were repeatedly stimulated with anti-CD3/CD28 Dynabeads every 3 to 4 days for 17 days (Figure 1A). Cell viability monitored longitudinally through the entirety of the protocol revealed no significant impact. Relative proliferation was found to be at maximal output on day 10 with a decline in subsequent rounds of stimulation (supplemental Figure 1A). Longitudinal flow cytometry analysis revealed a sustained increase in expression of exhaustion markers, such as thymocyte selection-associated HMG box (TOX), programmed cell death 1 (PD-1), T-cell immunoglobulin and mucin-domain containing 3 (TIM3), and ectonucleoside triphosphate diphosphohydrolase 1 (ENTPD1; CD39) within both CD4 and CD8 T-cell populations (Figure 1B-C; supplemental Figure 1B-D). Cytokine and chemokine panel analyses showed a gradual decrease in secretion, resulting in 62% decrease for interleukin 2 (IL-2), 94% for CCL17, 92% for MIP3a, and 58% for CCL5 by day 14 compared to day 3 measurements (Figure 1D).

To compare the functional capacity between T-cell states, we cocultured autologous activated T cells (Tact; 3 days after initial stimulation) or Tex (day 14 of chronic stimulation) with K562 target cells ectopically expressing BCMA in the presence of alnuctamab, a BCMA-CD3 TCE. Although there was a dose-dependent increase in alnuctamab-mediated T-cell killing, Tex exhibited a significantly reduced killing capacity of target cells compared to Tact, by both incucyte and flow analyses (Figure 1E-F; supplemental Figure 1E).

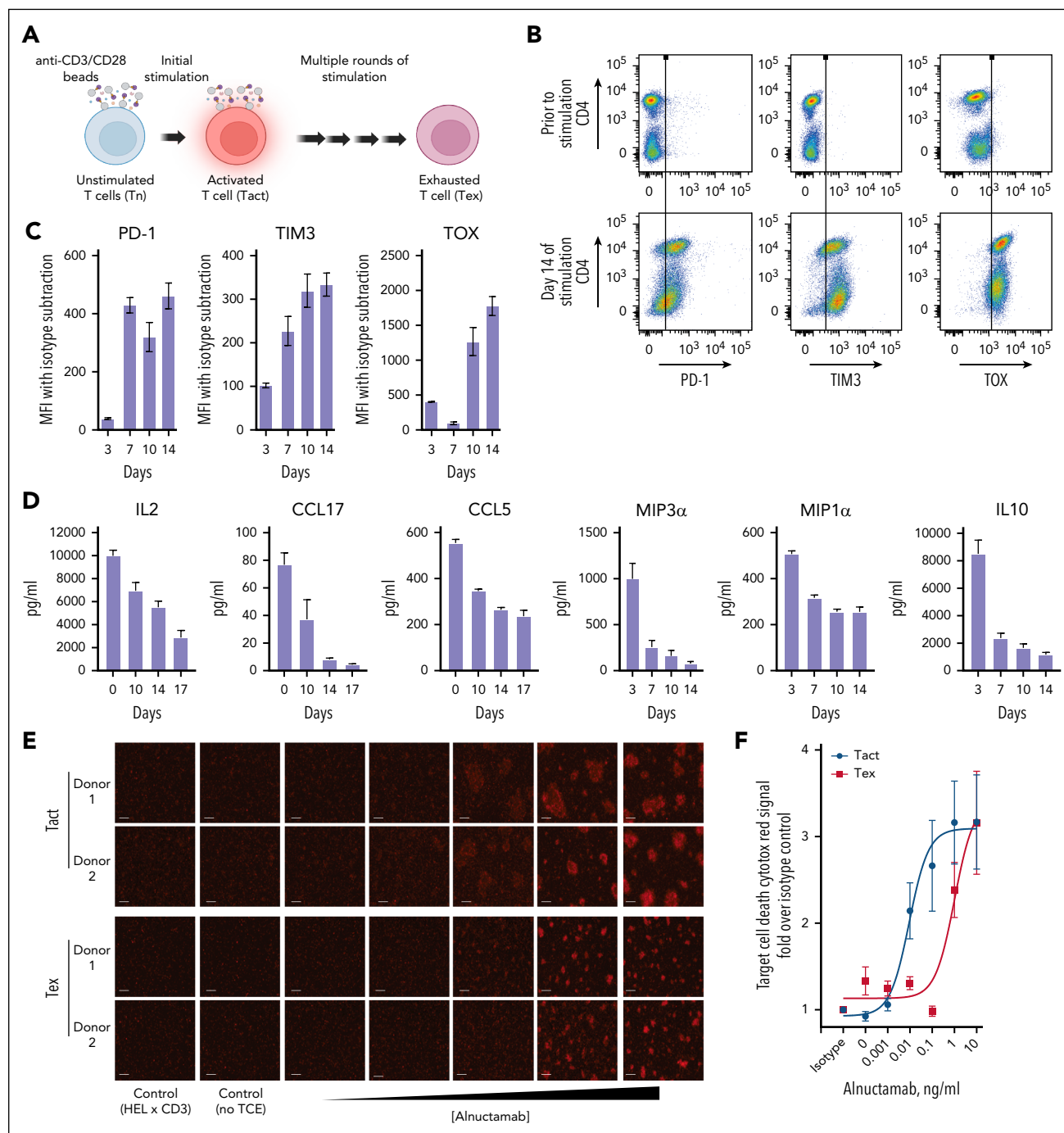


Figure 1. Confirmation of T-cell exhaustion from ex vivo generation of Tex. (A) Schematic of ex vivo generation of Tex using anti-CD3/CD28 Dynabeads starting from total T cells isolated from healthy donors. Figure created in BioRender. Chiu H. (2026) <https://biorender.com/k27tm08>. (B) Sample scatterplots from flow cytometry analysis showing upregulation of exhaustion markers TOX, PD-1, and TIM3 before stimulation and 14 days after multiple rounds of stimulation. (C) Longitudinal flow analysis on exhaustion markers TOX, PD-1, and TIM3. Data are presented as mean $\pm$ standard error of the mean (SEM) ($n \geq 4$). (D) Milliplex multiplex panel analysis showing a longitudinal reduction in cytokine and chemokine secretion. Data are presented as mean $\pm$ SEM ($n \geq 8$). (E) Representative images from Incucyte showing coculture assays of either Tact or Tex with BCMA expressing K562 target cell in the presence of increasing doses of alnuctamab (0.1, 1, 10, 100, and 1000 ng/mL), comparing potency of target cell killing capacity of Tact to Tex (left). Red fluorescence from Cytotox reagent-labeled dead target cells. Scale bar, 200 μ m. (F) Coculture killing assay analyzed using flow cytometry gating on apoptotic and dead target cells, comparing 50% inhibitory concentration of alnuctamab from Tact to Tex. Data are presented as mean $\pm$ SEM ($n = 4$). HEL, hen egg lysozyme; MFI, mean fluorescence intensity.

State-specific IKZF1 chromatin localization and bidirectional transcriptional regulation in T-cell activation and exhaustion

Given that T cells undergo significant changes in their nuclear architecture and chromatin accessibility during differentiation

or in the phenotypic stage, such as activation and exhaustion^{19,24} and a thorough genome-wide analysis of occupancy and transcriptional regulatory activity between Tex and Tact has not been performed, we aimed to investigate the chromatin binding patterns of IKZF1. We conducted an integrated

epigenomic investigation using Hi-C, ATAC-seq, RNA-seq, and ChIP-seq, focusing on IKZF1 and histone modifications, including H3K27ac, H3K27me3, H3K4me1, and H3K4me3. This approach allowed us to understand IKZF1-specific regulation of transcription by characterizing the epigenome in Tex vs Tact.

Initial analysis focused on identifying IKZF1 binding sites across the genome. We discovered 19 964 common peaks found between Tact and Tex, 16 774 peaks unique to Tact, and 14 759 peaks specific to Tex (Figure 2A). Notably, Tact-specific peaks were less prevalent in promoter regions (within 1 kb) compared to peaks specific to Tex or those shared between both conditions (Figure 2B). Examining the relationship between IKZF1 genomic localization and histone modifications associated with promoters (H3K4me3), enhancers (H3K4me1), and both (H3K27ac), we categorized the peaks based on their binding patterns in each state: shared, Tact specific, and Tex specific. Tact-specific IKZF1 peaks were markedly reduced in Tex, whereas Tex-specific peaks exhibited diminished enrichment in Tact, signifying a reassignment of IKZF1 binding between activation and exhaustion states. We then assessed the colocalization of these IKZF1-bound regions with histone modifications. When comparing exhaustion to the activation state, calculated by area under the curve of the average enrichment profiles for each histone mark, we observed an 11% reduction from Tact to Tex at Tact-specific peaks for H3K27ac, whereas Tex-specific peaks showed a 30% increase from Tact to Tex. For H3K4me1, we observed a 25% decrease at Tact-specific peaks and a 39% increase at Tex-specific peaks. For H3K4me3, the changes were more modest, with a 4% decrease at Tact-specific peaks and a 10% increase at Tex-specific peaks (Figure 2C). Notably, chromatin accessibility at IKZF1-defined loci across all 3 categories in Tex showed decreased signal intensity (supplemental Figure 2A).

Gene set enrichment analysis (GSEA) of proximal genes within 3 kb to IKZF1 peaks revealed distinct functional associations. Tact-specific IKZF1 binding sites were enriched for close genes involved in cytokine signaling pathways, including IL-2/STAT5 and IL-6/JAK/STAT3 pathways (supplemental Figure 2B). In contrast, Tex-specific IKZF1 binding sites were significantly associated with genes involved in pathways that are canonically upregulated in Tex compared to other T cell subsets, including memory, effector, and naive T cells (supplemental Figure 2C). To further elucidate the differences between activation and exhaustion states, we performed a comparative analysis of chromatin accessibility and mRNA abundance; 2124 peaks were identified with significantly altered chromatin accessibility, and 1359 genes with altered gene expression, between the 2 states (supplemental Figure 2D-E). Interestingly, the peaks with gained chromatin accessibility also showed increased IKZF1 binding signal, whereas those with lost chromatin accessibility were accompanied by reduced IKZF1 binding (Figure 2D). Confirming prior observations associated with IKZF1 binding, cytokine-related genes such as IL-1R, IL-17A, and IL-2 exhibited decreased chromatin accessibility and mRNA abundance in Tex compared to Tact. Conversely, genes associated with T-cell inhibitory functions, including ENTPD1 and TOX, were upregulated in Tex compared to Tact.

Next, to map the transcriptional regulation underlying the transition from Tact to Tex, we used the activity-by-contact (ABC) model for enhancer gene prediction, which calculates an ABC score based on the activity of an enhancer and its contact frequency with a gene's promoter. Enhancer activity is determined by the read counts of ATAC-seq and H3K27ac ChIP-seq at the enhancer, whereas contact frequency comes from the Hi-C data, showing how physically close the enhancer and the promoter are in the 3 dimensional genome.²⁵ In our ABC analysis, ~65 000 active enhancer-promoter (E-P) interactions were identified in both Tact and Tex samples. Comparing Tex to Tact, ~17 000 E-P interactions were active in Tex but not in Tact, and ~19 000 active in Tact but not in Tex (Figure 2E). For each gene, we calculated a gene-level ABC score by adding up the scores of all E-P interactions at its promoter, then computed the score difference between Tex and Tact to identify genes most influenced by E-P interaction dynamics (Figure 2F). Comparing genes with the greatest gains and losses in E-P interactions (absolute score change >0.2), we identified 400 genes with gained E-P and 363 genes with reduced E-P, which showed a significant difference in expression ($P = 2.3 \times 10^{-6}$; Figure 2G). Additionally, we noted consistent changes in IKZF1 binding at the enhancer regions of gained or reduced E-P interactions (higher binding at enhancers of gained E-P interactions and lower binding at those lost) suggesting that IKZF1 may play a role in modulating enhancer activity and its interactions with promoters (Figure 2H).

These data highlight the important and distinct role IKZF1 exerts in shaping the transcriptional regulatory landscape and downstream gene expression profiles characteristic of both T-cell activation and exhaustion.

Epigenomic-driven remodeling distinguishes the chromatin architectural landscape between activated cells and Tex

To understand the dynamic changes in DNA architecture associated with gene expression in Tex, we used Hi-C data to analyze the 3-dimensional genome organization in each T-cell state. Compared to Tact, Hi-C analysis showed an increase in compartment-level interactions (≥ 10 Mb) in Tex, whereas interactions related to topologically associated domains (TADs) and loops were decreased (< 10 Mb) (supplemental Figure 3A). We identified chromatin compartments, broadly categorized into compartment A (active) and compartment B (inactive), where compartment A is defined by regions typically exhibiting higher transcriptional activity. When comparing Tex vs Tact, ~150 genomic bins (defined by window of 100 kb) transitioned from compartment A to compartment B, ~200 switched from compartment B to A, whereas ~1200 bins remained stable (Figure 3A). Probing the connection of epigenetic states that accompanied compartment changes revealed increased deposition of H3K4me1, H3K4me3, and H3K27ac in genomic loci that switched from compartment B in Tact to A in Tex, indicating a potential for increased transcription (Figure 3B). IKZF1 binding was also found to be stronger in these regions, suggesting that IKZF1 plays a role in regulating the expression of these genes (Figure 3B). Interrogation of RNA-seq indeed confirmed that genes located within regions that transitioned

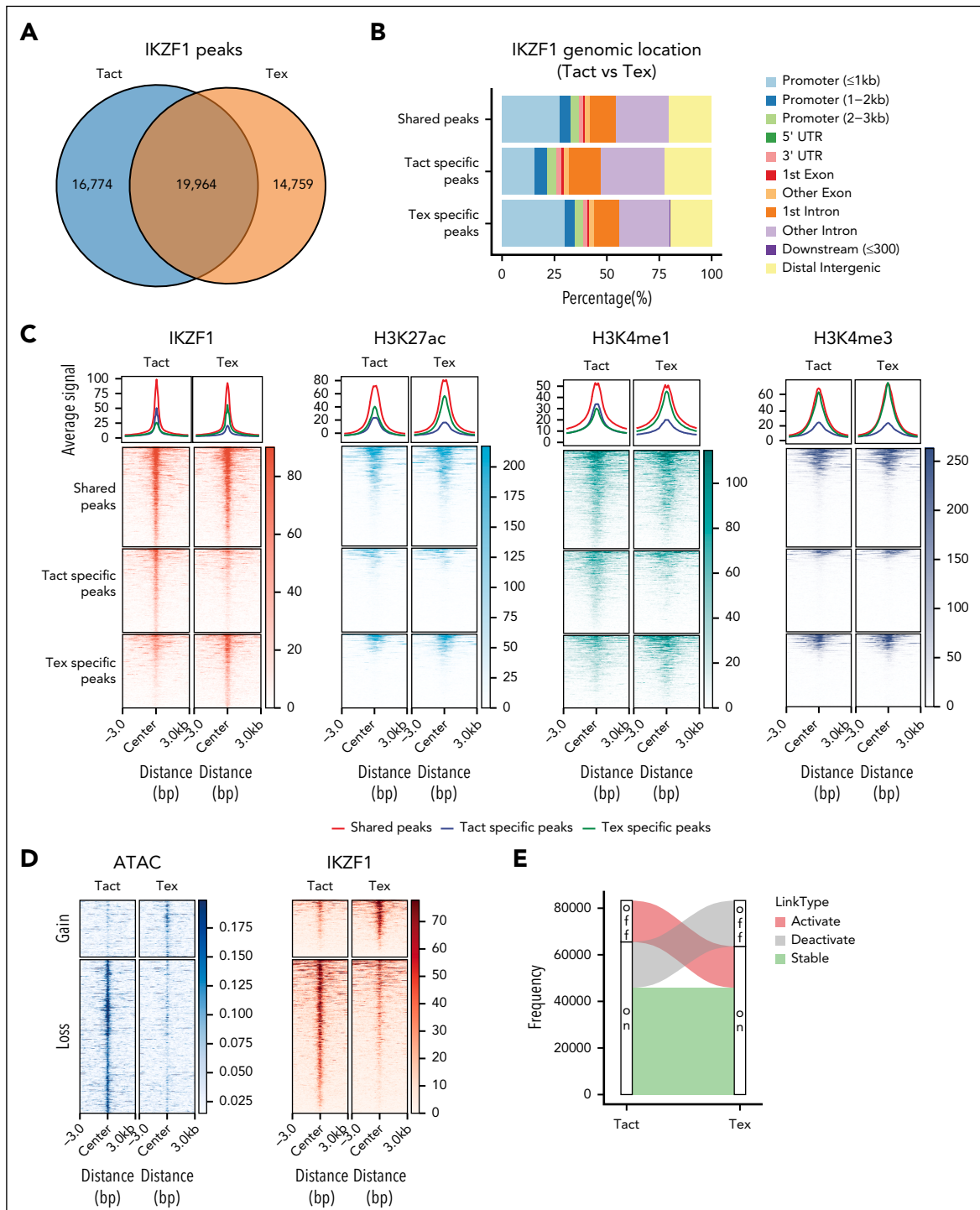


Figure 2. IKZF1 is involved in transcriptional regulatory chromatin interactions during transition from Tact to Tex. (A) Venn diagram showing the overlap of IKZF1 peaks in Tact and Tex. (B) Annotation of IKZF1 binding sites divided into 3 categories: shared by Tact and Tex, Tact specific, and Tex specific. (C) Heat map for ChIP-seq showing the enrichment of H3K27ac, H3K4me1, and H3K4me3 signal in IKZF1 peaks defined in panel A. (D) Left: heat map illustrating the differential ATAC peaks categorized as gain and loss in Tex compared to Tact. Right: corresponding heat map showing IKZF1 binding at these differential ATAC peaks. (E) Alluvial plots showing the dynamic changes of E-P interactions during Tact transition to Tex. The number of E-Ps with activation (OFF to ON, red) or deactivation (ON to OFF, gray) between conditions is indicated by the thickness of the lines. (F) Waterfall plot showing the ABC score difference between Tex and Tact for each gene. Genes with a positive ABC score difference and upregulated in Tex are colored red, whereas those with a negative score and downregulated in Tex are colored blue. Two genes, *ENTPD1* and *IL1R1*, are highlighted in the plot. (G) Box plot comparing the log₂ fold change (Tex/Tact) of genes with reduced E-P interactions (negative ABC score difference in Tex compared to Tact) vs genes with gained E-P interactions (positive ABC score difference). Statistical analysis was performed using an unpaired 2-tailed Student t test. (H) IKZF1 binding signal intensity at the enhancer side and promoter side for the reduced and gained E-P in Tex compared to Tact. UTR, untranslated region.

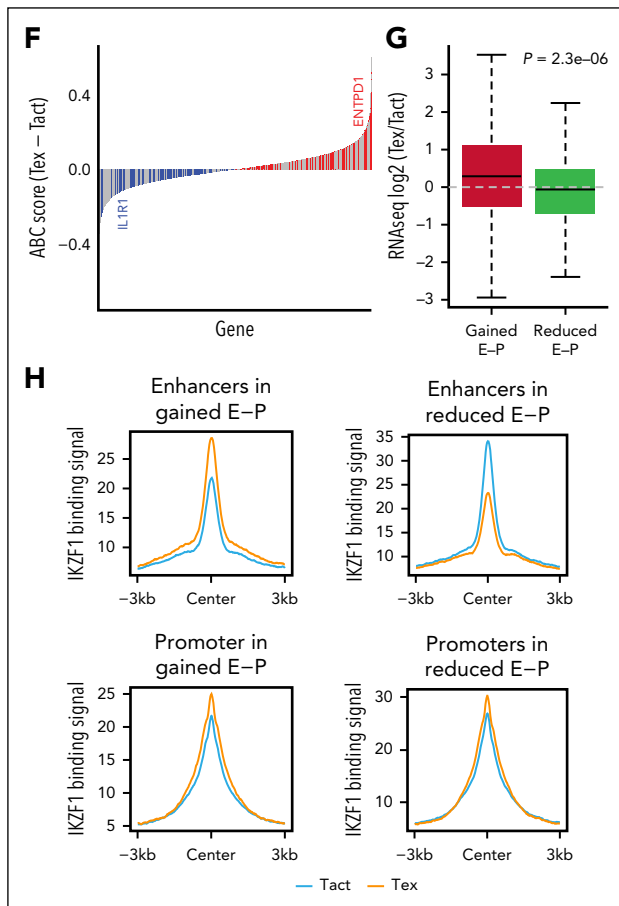


Figure 2 (continued)

from compartment B to A showed increased mRNA expression (Figure 3C).

Focusing on exhaustion markers, we found that *ENTPD1* was among the top genes with increased E-P interactions observed in Tex (Figure 2F). *ENTPD1* has been shown to mark terminal Tex, and its expression directly correlated to reduced T-cell function and cancer prognostic.²⁶⁻²⁸ To gain deeper insight into the mechanisms driving this observation, we first analyzed Hi-C data to compare 3-dimensional genome organization across chromosome 10 (80-100 Mb), where *ENTPD1* is located, in each T-cell state. Our analysis revealed that the DNA region encompassing *ENTPD1* and its neighboring genes exhibited a negative principal component 1 (PC1) value in Tact (Figure 3D), indicating classification in compartment B. However, this region showed a positive PC1 value in Tex, reflecting a shift toward compartment A, a more active chromatin state. This observation suggested that the transcriptional upregulation of *ENTPD1* in Tex is accompanied by changes in the local chromatin architecture, potentially facilitating its increased expression. Next, we examined TADs, which are large, self-interacting genomic regions that regulate gene expression by facilitating or restricting interactions between promoters and enhancers around *ENTPD1*, to investigate the specific local changes in DNA conformation. As indicated by the change in ABC scores, we observed increased TAD strength encompassing *ENTPD1* in Tex, characterized by a differential increase in short-range intrachromosomal interactions compared to

Tact (Figure 3E-F; supplemental Figure 3E). This suggested enhanced DNA connectivity between promoters and enhancers of *ENTPD1*, as defined by H3K27ac marks, in the exhausted state. We then integrated ChIP-seq data for IKZF1, H3K27ac, H3K4me1, and H3K4me3 with ATAC-seq and RNA-seq profiles (Figure 3G). Overlaying these data revealed an increased enrichment of IKZF1 at promoter and enhancer regions in Tex compared to Tact. The augmented presence of IKZF1, in conjunction with the active histone modifications H3K27ac and H3K4me3, as well as the enhancer-associated mark H3K4me1, indicates a coordinated regulation of chromatin accessibility and gene expression in the exhausted state. Likewise, this coordinated regulation of gene expression is also evident at the *PDCD1* locus, which encodes PD1. Quantitatively, *PDCD1* expression increased by 17% from Tact to Tex, with concordant increases in IKZF1 binding, histone modifications (H3K27ac, H3K4me1, H3K4me3), and chromatin accessibility at the *PDCD1* promoter and nearby enhancers, with all signals shifting in the same direction as *ENTPD1* (supplemental Figure 3B-D).

Conversely, *IL1R1*, a cytokine-related gene whose expression decreased from activated to exhausted state (supplemental Figure 2D-E), is among those with most reduced E-P interactions in Tex (Figure 2F). Upon examination of *IL1R1*, we observed a shift toward the B compartment, reflected by a decreased PC1 value at the *IL1R1* locus, along with diminished formation of TAD encompassing this region in Tex (Figure 3H-J; supplemental Figure 3F). This reduction in TAD strength indicates decreased interactions between promoters and enhancers within the domain, which is consistent with the results observed from our ABC model analysis. Additionally, we observed a loss of IKZF1 binding at both enhancer and promoter regions of *IL1R1* in Tex. This was corroborated by a decrease in active histone marks at these regulatory regions (Figure 3K). Similarly, *IL17F*, another downregulated cytokine-related gene, exhibited reduced TAD strength and diminished IKZF1 binding at its enhancer regions in Tex, accompanied by decreased active histone marks (supplemental Figure 3G-I).

These cumulative findings indicate that IKZF1 redistribution plays a role in shaping the transcriptional program associated with T-cell exhaustion by either promoting or repressing gene expression through its effects on the epigenetic landscape.

Mezigidomide-mediated IKZF1 and IKZF3 degradation alters the chromatin landscape, resulting in increased cytokine expression and reduction in exhaustion markers in Tex

To better elucidate the transcriptomic programming specifically regulated by IKZF1, Tex (day 14 after initial stimulation) were treated with either dimethyl sulfoxide (DMSO) or mezigidomide for another 3 days. Both IKZF1 and IKZF3 were confirmed to be degraded in treated samples, with upregulation of proinflammatory cytokine and chemokine secretion (supplemental Figure 4A-B). Integrated epigenomic and transcriptomic data were generated to determine whether mezigidomide treatment influences chromatin architecture, accessibility, and gene expression.

Global distribution of the IKZF1 ChIP-seq signal revealed a significant reduction in mezigidomide-treated cells compared

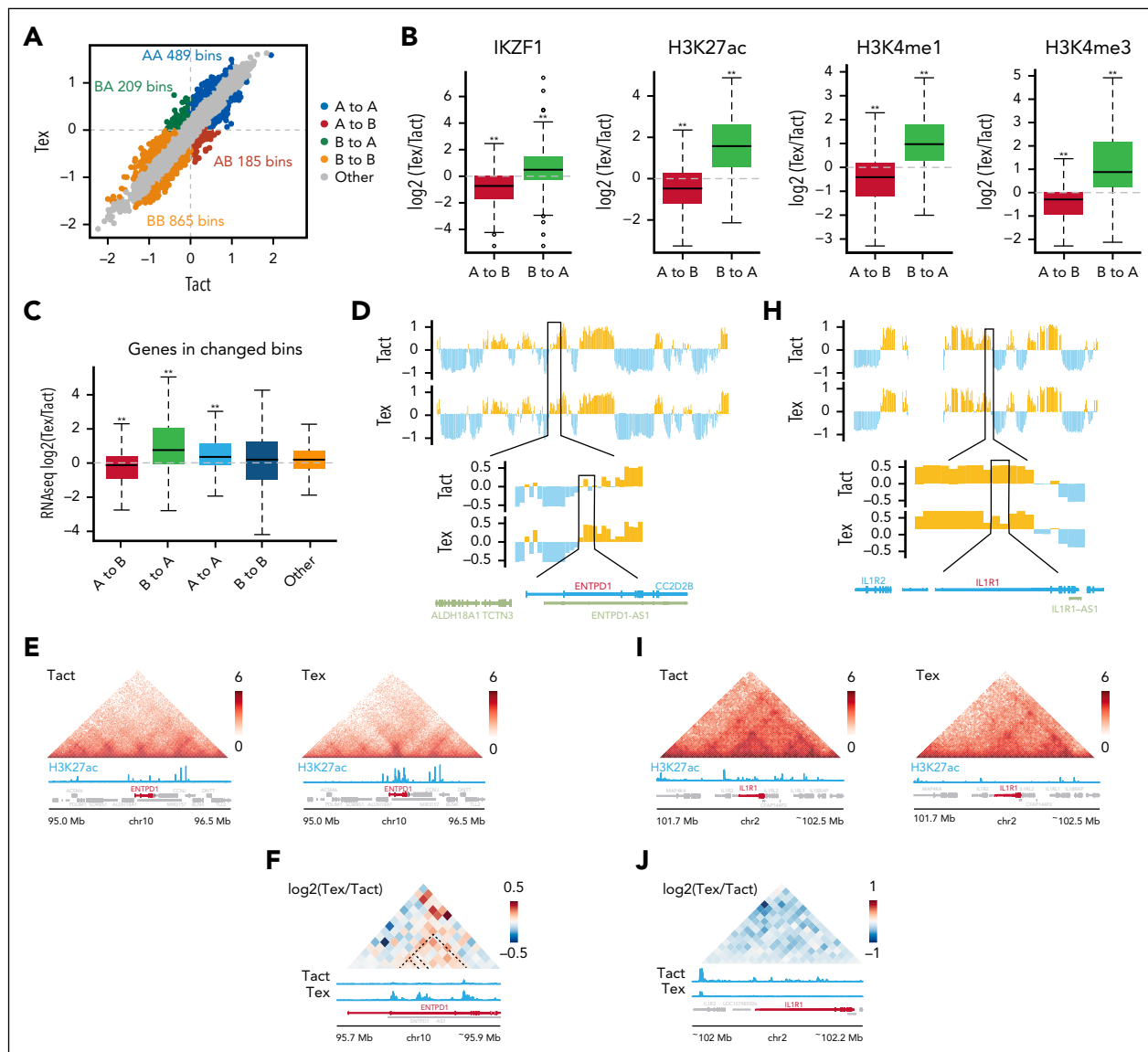


Figure 3. Chromatin compartment changes associated with T-cell exhaustion. (A) PC1 of Tact and Tex from Hi-C compartment analysis. The number of significantly changed 100 kb bins is shown in each quadrant. Significantly changed bins were identified by calculating the absolute value difference between the PC1 of Tact and Tex, with a threshold of ≥ 0.2 considered significant. (B) Difference in aggregate signal levels of IKZF1, H3K27ac, H3K4me1, and H3K4me3 ($n = 3$) within significantly switched bins identified in panel A. (C) Differential expression ($n = 3$) of genes associated with each category identified in panel A from RNA-seq analysis. Significance was determined using 1-sample 2-tailed t test. (D) Chromatin state of the *ENTPD1* locus comparing Tex with Tact. Each track shows Hi-C compartment PC1, with orange indicating compartment A and blue indicating compartment B. (E) Top: Hi-C plots of the region surrounding the *ENTPD1* gene in Tact and Tex. (F) Difference heat map showing the region surrounding the *ENTPD1* gene with increased (red) and decreased interactions (blue) in Tex. (G) Normalized ChIP-seq, ATAC-seq, and RNA-seq tracks of the *ENTPD1* locus. (H) Chromatin state of the *IL1R1* locus comparing Tex with Tact. Each track shows Hi-C compartment PC1, with orange indicating compartment A and blue indicating compartment B. (I) Similar layout as panel E for the *IL1R1* locus, Hi-C plots of the region surrounding the *IL1R1* gene in Tact and Tex. (J) Difference heat map showing the region surrounding the *IL1R1* gene with increased (red) and decreased interactions (blue) in Tex. (K) Normalized ChIP-seq, ATAC-seq, and RNA-seq tracks of the *IL1R1* locus. $**P < .01$.

to vehicle control, confirming the effective degradation of chromatin-bound IKZF1 (Figure 4A). Global ATAC-seq analysis revealed a substantial increase in chromatin accessibility in Tex treated with mezigdomide (Figure 4B; supplemental Figure 4C). Motif enrichment analysis showed significant enrichment of transcription factors associated with Tact such as NFATC1/2/3^{29,30} in the newly opened chromatin after mezigdomide treatment (Figure 4C). RNA-seq analysis confirmed that mezigdomide treatment led to bidirectional transcriptional changes with both upregulated and downregulated genes

(supplemental Figure 4D). Notably, we observed an upregulation of cytokine and cytokine-related genes, such as CCL17, which was repressed during the transition from activation to exhaustion (supplemental Figure 2E). Consistent with this observation, GSEA showed that pathways related to immune response were significantly activated after mezigdomide treatment (supplemental Figure 4E).

Investigating DNA conformation change through Hi-C data set in Tex, we found a slight decrease in compartment-level

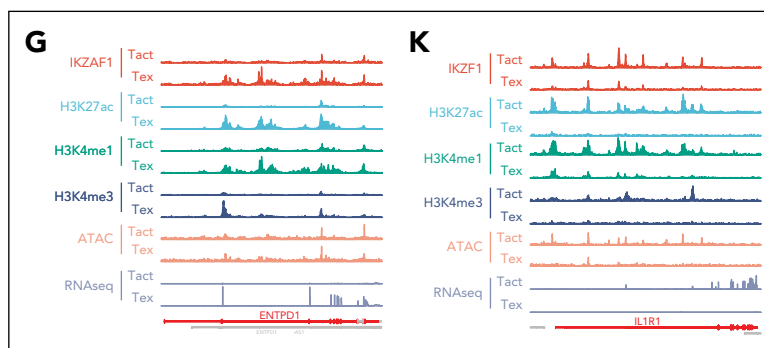


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interactions (≥ 10 Mb) following mezigdomide treatment (supplemental Figure 5A). AB compartment analysis revealed transition of 53 bins from B to A compartments, and 30 bins from A to B (supplemental Figure 5B). These compartmental shifts were accompanied by alterations in histone modifications, as assessed by ChIP-seq, and significant changes in gene expression, as confirmed by RNA-seq analysis (supplemental Figure 5C-D). These findings suggest that mezigdomide-mediated degradation of IKZF1 and IKZF3 induces changes in chromatin architecture, potentially influencing gene regulation through modifications in both DNA conformation and histone mark depositions.

To evaluate the functional impact of chromatin changes on gene regulation, we applied the ABC model to compare mezigdomide-treated Tex with the DMSO control and found ~19 000 active E-P interactions exclusively in the mezigdomide-treated Tex vs ~17 000 in the DMSO group (Figure 4D). This increase in E-P interactions was linked to a notable upregulation of gene expression after mezigdomide treatment (Figure 4E-F). Focusing on the E-P interactions exclusively activated in mezigdomide-treated Tex, we observed greater IKZF1 binding enrichment at the enhancers compared to the promoters (Figure 4G). This pattern further suggests that IKZF1-mediated transcriptional regulation may, at least in part, occur through distal enhancers rather than directly at the promoters.

Comparing mezigdomide-treated Tex to vehicle control, CCL1 and 2 were among the top genes showing increased E-P interactions in the treated cells (Figure 4E). Not surprisingly, further examination of the Hi-C data revealed a strengthening of TAD encompassing CCL1 and 2 in Tex after mezigdomide treatment (Figure 5A-B; supplemental Figure 5E). Integration of ChIP-seq and RNA-seq data demonstrated that mezigdomide treatment reduced promoter-bound IKZF1 and led to the emergence of an intergenic enhancer marked by H3K4me1 and H3K27ac (Figure 5C). This postmezigdomide enhancer is correlated with increased secretion of CCL1 and 2 (supplemental Figure 4B).

Conversely, mezigdomide treatment in Tex resulted in a slight diminution of the TAD strength surrounding *PDCD1* (Figure 5D-E; supplemental Figure 5F). Visualization of the integrated ChIP-seq and RNA-seq tracks revealed reduced IKZF1 binding at both promoter and enhancer regions after mezigdomide treatment. This reduction in IKZF1 binding correlated with decreased enhancer activity, as indicated by

reduced H3K27ac and H3K4me1 marks at upstream and downstream enhancers of *PDCD1* and decreased RNA expression (Figure 5F).

Single-cell multiomic analysis confirms mezigdomide reversal of exhaustion-related gene expression in Tex of patients with primary MM

To validate our preclinical findings, we investigated the response to mezigdomide treatment in Tex from patients with MM. MM PBMCs were stimulated with anti-CD3/CD28 Dyna-beads and treated with either DMSO or mezigdomide for 3 days, followed by cytokine expression analysis and PBMCs to perform dual scRNA-seq and scATAC-seq within the same cells to capture the differential molecular alterations. Uniform manifold approximation and projection visualization of the scRNA-seq data revealed distinct clustering of major immune cell populations, including B cells, T cells, natural killer cells, and Tregs, allowing for precise identification and subsequent analysis of these cell types (Figure 6A). Within this uniform manifold approximation and projection, we further identified ENTDP1-positive T cells to delineate the Tex population.

Focusing our analysis on ENTDP1-positive T cells, we examined the expression and chromatin accessibility of key cytokine genes, including interferon gamma (IFN- γ), CCL17, and CCL2. Single-cell analysis demonstrated that mezigdomide treatment led to a significant increase in both gene expression and chromatin accessibility (Figure 6B). Furthermore, a significantly strong correlation between gene expression and accessibility at nearby enhancer peaks was observed, suggesting a mechanistic basis for the enhanced gene expression in mezigdomide-treated cells. GSEA of the differentially expressed genes in ENTDP1+ Tex revealed significant enrichment of pathways related to tumor necrosis factor α signaling by NF- κ B, IFN- γ response, and IFN- α response in the mezigdomide-treated group (Figure 6C). These findings have been confirmed by cytokine release (Figure 6D), demonstrating the role of IKZF1 in modulating key signaling pathways associated with T-cell function in patients with MM.

Mezigdomide treatment results in reinvigoration of Tex in killing tumor cells with TCE

To determine whether functional impairment of Tex could be reversed through mezigdomide treatment, Tex were cocultured with BCMA-expressing K562 cells and treated with alnuctamab, followed by treatment with either DMSO or mezigdomide for another 6 days. Mezigdomide-treated Tex

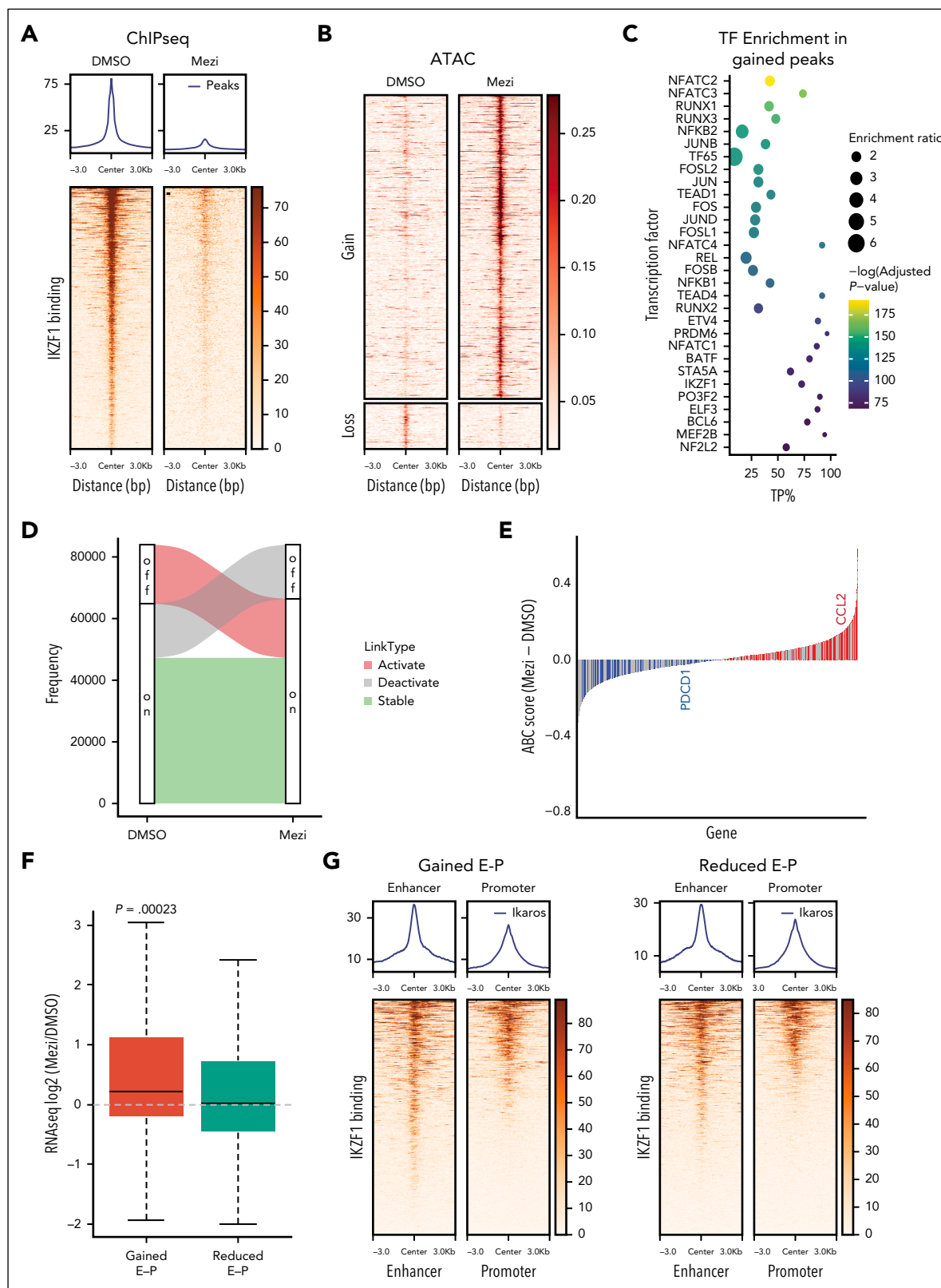


Figure 4. Mezigdomide (Mezi) treatment alters transcriptional regulation by modulating enhancer activity in Tex. (A) Heat map of IKZF1 binding signal in Tex before and after Mezi treatment. (B) Heat map depicting the differential ATAC peaks classified as gain and loss in Tex following Mezi treatment. (C) Dot plot showing TF enrichment analysis of gained open ATAC-seq peaks identified in panel B. TP% on the x-axis label refers to the percentage of true positive peaks that contain the given motif. The color of the dot reflects the adjusted *P* value, and its size corresponds to the relative enrichment ratio. (D) Alluvial plots showing the dynamic changes of E-P interactions after Mezi treatment in Tex. The number of E-Ps with activation (OFF to ON, red) or deactivation (ON to OFF, gray) between conditions is indicated by the thickness of the lines. (E) Waterfall plot showing the ABC score difference between Mezi-treated Tex and DMSO control for each gene. Genes with a positive ABC score difference and upregulated after Mezi treatment are colored red, whereas those with a negative score and downregulated are colored blue. Two genes, *CCL1* and *PDCD1*,

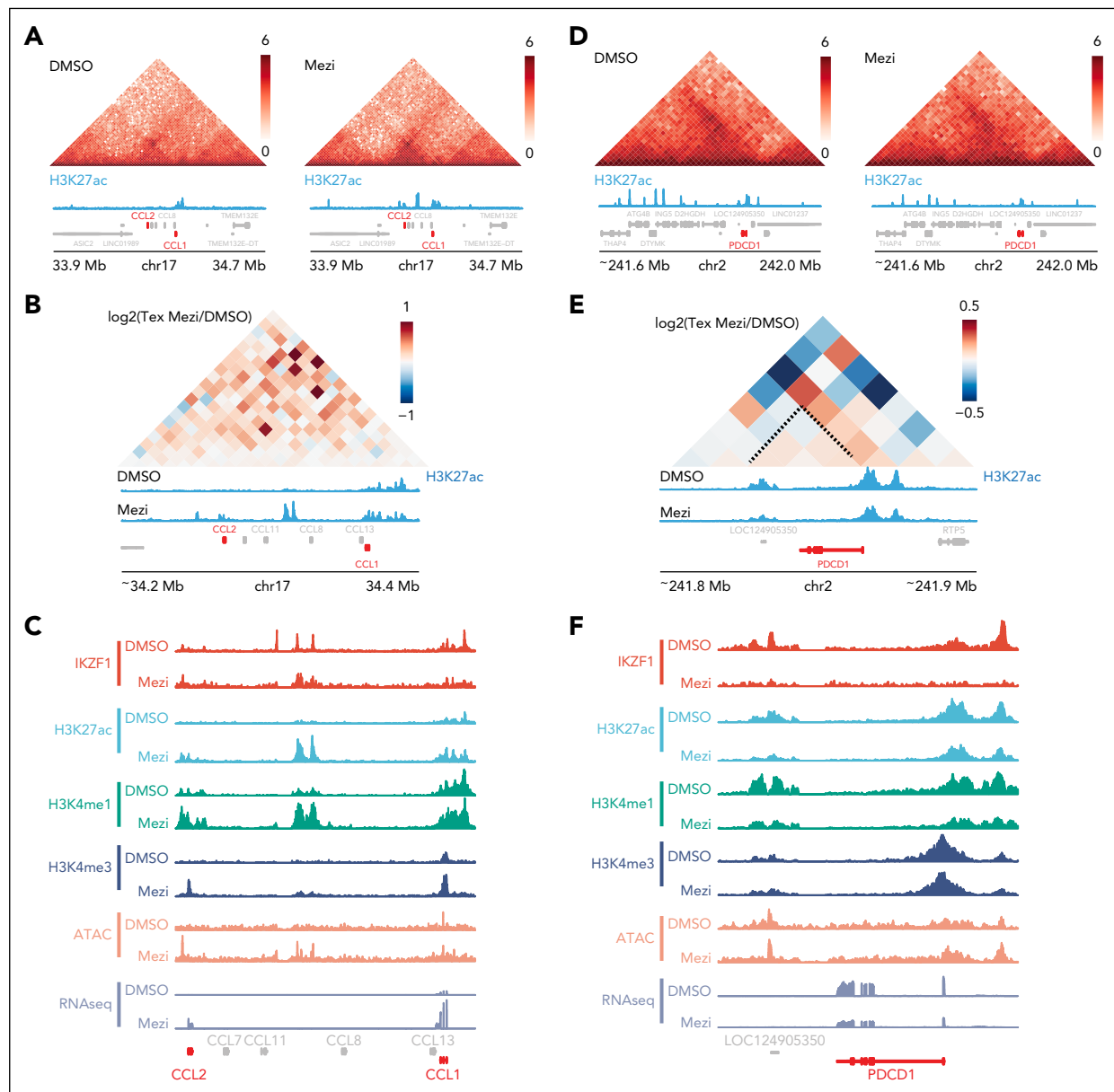


Figure 5. Chromatin state and interaction dynamics at CCL1 and 2 and PDCD1 loci in Tex after Mezi treatment. (A) Top: Hi-C plots of the region surrounding the CCL1/2 genes in Mezi-treated Tex and DMSO control. (B) Difference heat map showing the region surrounding the CCL1/2 gene with increased (red) and decreased (blue) interactions in Mezi-treated Tex. (C) Normalized ChIP-seq and RNA-seq tracks of the CCL1 and 2 loci. (D-F) Similar layout as panels A-C for the *PDCD1* locus.

exhibited over fourfold increase in the killing of target cells compared to alnuctamab alone (Figure 7A-B), reducing 50% effective concentration of alnuctamab to levels comparable to Tact (supplemental Figure 6A). Flow cytometry analysis of Tex 3 days after coculture revealed that alnuctamab dose-dependently increased the expression of IKZF1, IKZF3, PD1, and TIM3 and activation-associated markers RUNX3 and CD40L, whereas mezigdomide treatment significantly down-regulated the exhaustion markers and upregulated the activation markers (supplemental Figure 6B).

To further demonstrate the broad applicability of our results, we used 2 additional clinically approved TCEs, talquetamab and teclistamab, with cereblon-null H929 pomalidomide-resistant and MM1S pomalidomide-resistant cells as target cells.³¹ As shown in supplemental Figure 6C-D, mezigdomide enhanced Tex cytotoxicity against these target cells in combination with the additional TCE molecules. Furthermore, this enhanced cytotoxicity was also observed in PBMCs of patients with MM in combination with elranatamab (supplemental Figure 6E).

Figure 4 (continued) are highlighted in the plot. (F) Box plot comparing the log₂ fold change (Mezi/DMSO) of genes with reduced E-P interactions (negative ABC score difference in Mezi-treated Tex compared to DMSO) vs genes with gained E-P interactions. (G) IKZF1 binding signal intensity at the enhancer side and promoter side for the gained and reduced E-P in Mezi-treated Tex compared to DMSO. TF, transcription factor.

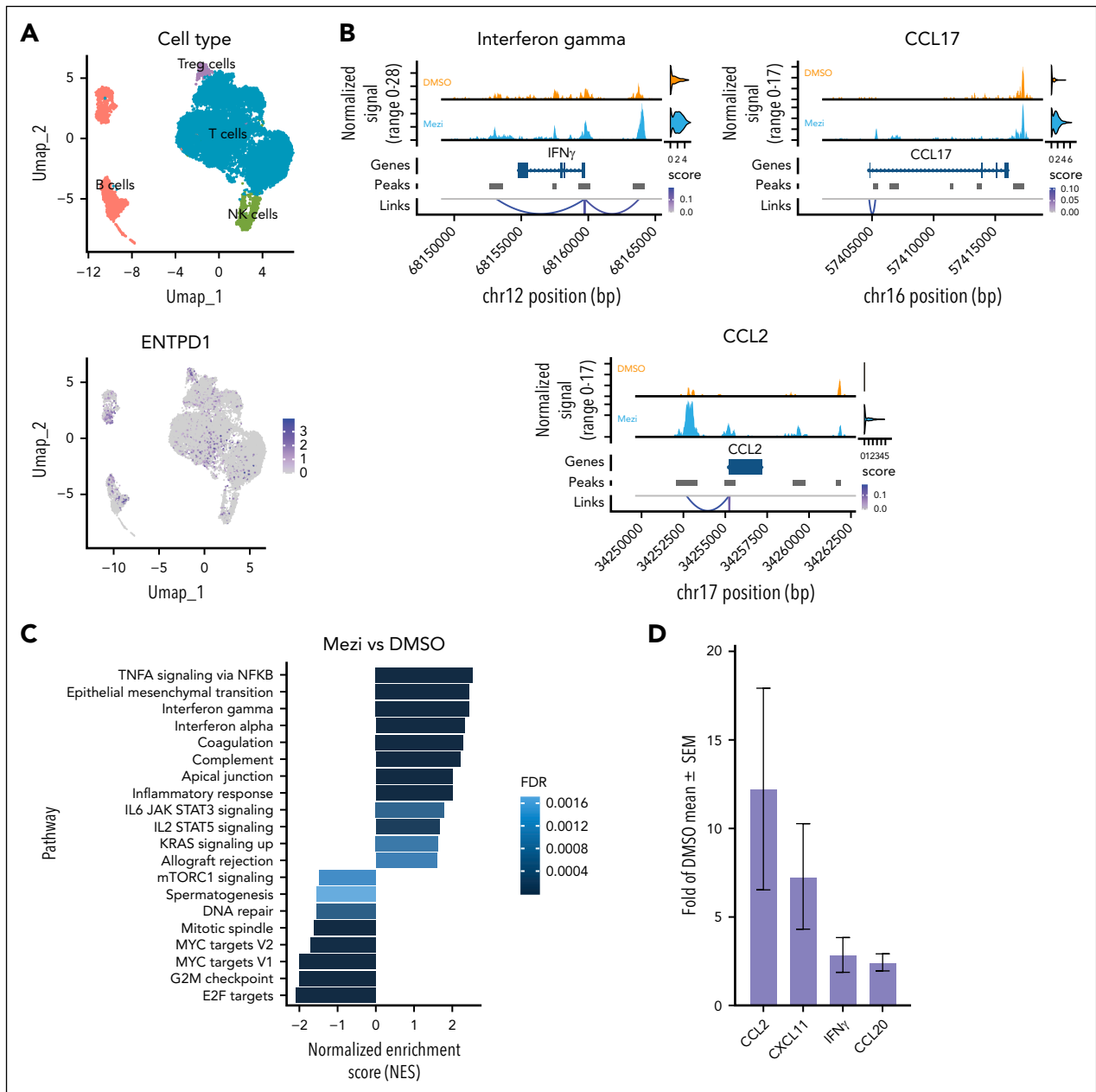


Figure 6. Single-cell multiome ATAC + gene expression sequencing analysis of PBMCs from patients with MM reveals response to Mezi treatment. (A) UMAP plot of PBMCs from patients with MM (n = 4). Top: cell type annotations for each cluster are shown. Bottom: expression levels of ENTPD1 across the same UMAP. (B) Coverage plot of IFN- γ , CCL2, and CCL17 genes using Signac in ENTPD1+ T cells from patients with MM comparing Mezi treatment to DMSO control. (C) Hallmark pathway enrichment analysis showing upregulated and downregulated pathways in Mezi-treated ENTPD1+ T cells compared to DMSO control. (D) Cytokine analysis using supernatants obtained from MM PBMCs (n = 4 different donors) stimulated with anti-CD3/CD28 Dynabeads and treated with Mezi (1nM) or DMSO for 3 days. Data are normalized to DMSO controls. FDR, false discovery rate; UMAP, uniform manifold approximation and projection.

Together, these data demonstrate that mezigdomide-mediated IKZF1 degradation restores Tex function, leading to a greater cytotoxic capacity in combination with a TCE.

Discussion

In this study, we aimed to elucidate the epigenetic and transcriptomic underpinnings of T-cell exhaustion, which presents a significant clinical challenge in patients with relapsed and refractory MM receiving TCE, with a specific focus on the role of IKZF1 in the contribution to and maintenance of Tex.

Through integrated analyses of epigenetic and transcriptomic data sets comparing Tex to Tact, our findings indicated that IKZF1 predominantly exerts its regulatory effects on gene expression at both promoter and enhancer level, leading to increased expression of exhaustion genes but decreased expression of cytokine genes. This epigenetic reprogramming contributes to the functional decline, as demonstrated by the reduced efficacy of a BCMA TCE in a coculture model. The introduction of mezigdomide through the potent and specific degradation of IKZF1 effectively reversed the epigenetic and transcriptomic underpinnings of this exhaustion phenotype,

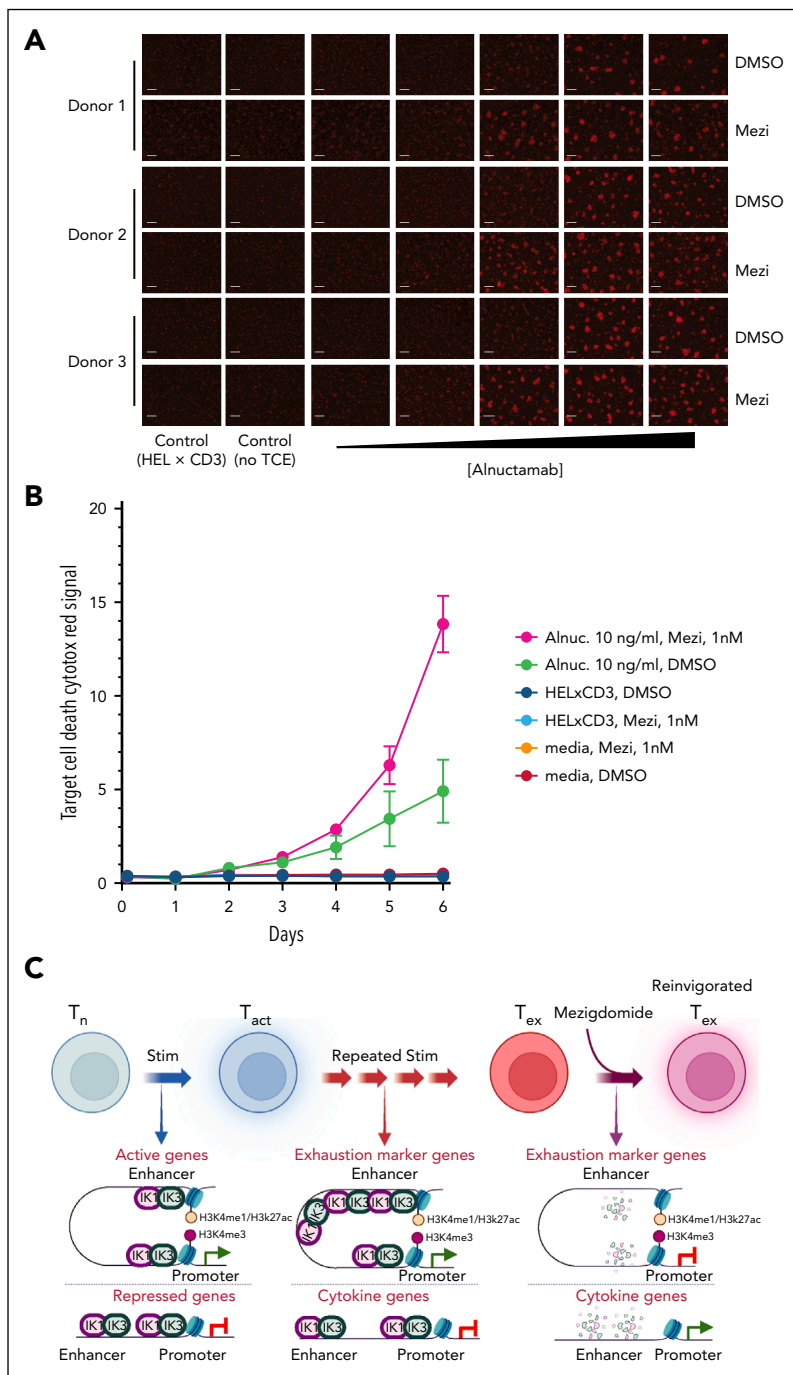


Figure 7. Mezi enhances the cytotoxic capacity of Tex in combination with alnuctamab. (A) Representative images from Incucyte with Tex cocultured with Cytotox (Sartorius)-labeled, BCMA-expressing K562 target cells. Each column is a titration of alnuctamab (0.1, 1, 10, 100, and 1000 ng/mL) treated with either DMSO or Mezi (1nM) for 6 days. Control isotype antibodies (HEL × CD3) were used at 1000 ng/mL. Red fluorescence from Cytotox reagent-labeled dead target cells. Scale bar, 200 μ m. (B) Longitudinal Incucyte analysis comparing the effect of Mezi (1nM) in Tex killing of target cells from coculture with 10 ng/mL of alnuctamab. Data are presented as mean $\pm$ SEM from 4 donors. (C) The presented model elucidates how IKZF1 and IKZF3 contribute to the epigenetic mechanisms underlying exhaustion to the development and maintenance of Tex. They bind to both enhancers and promoters of various genes. The binding of IKZF1 to exhaustion-related genes results in increased deposition of active chromatin marks (H3K4me1 and H3K27ac) and subsequent transcription of these genes. Mezi-mediated degradation of IKZF1/IKZF3 in Tex results in significant epigenetic reprogramming by reestablishing active chromatin marks at the cytokine gene loci, thereby restoring their expression while diminishing exhaustion gene expression. IK1, IKZF1; IK3, IKZF3. Figure created in BioRender. Chiu H. (2026) <https://biorender.com/thzelo5>.

culminating in enhanced TCE activity in Tex compared to TCE treatment alone (Figure 7C).

Interestingly, we observed that IKZF1 and IKZF3 protein levels increased with each round of stimulation, reaching their highest level in Tex (supplemental Figure 7A-B). This observation led us to hypothesize that mezigdomide, given the molecule's rapid and deep degradation of IKZF1 and IKZF3, may be particularly necessary in this setting due to this increased pool of cereblon substrates. Flow cytometry analysis indicated that mezigdomide could attain deeper levels of

degradation compared to lenalidomide and pomalidomide in this Tex setting (supplemental Figure 7C). This translated into increased production of IFN- γ and CCL17 (supplemental Figure 7D). Subsequent ChIP-polymerase chain reaction analysis revealed that mezigdomide alone could reduce IKZF1 occupancy at an enhancer proximal to IFN- γ in Tex (supplemental Figure 7E). Increased cytokine production and reversal of T-cell exhaustion led to increased activation status, as measured by HLA-DR and increased cytotoxicity in a coculture model utilizing elranatamab (supplemental Figure 7F-G).

IKZF1 was recently identified as a driver of T-cell exhaustion, and its degradation mediated by iberdomide prevented this process.³² In our study, we chose to utilize mezigdomide given the molecule's ability to completely degrade Ikaros and because the development strategy includes combinations with T-cell engagers. Iberdomide, while mechanistically related, is not currently an approved agent, and the registration clinical development program does not focus on the combination with TCE but, rather, on CD38 antibodies, such as daratumumab.

Importantly, our preclinical findings are corroborated by data presented in a companion clinical study that analyzed CYTOF profiles from bone marrow aspirates of patients treated with mezigdomide.³³ Their analysis revealed a consistent down-regulation of key exhaustion markers, including PD-1 and TIGIT, on T cells following treatment and provided translational validation of our mechanistic model, supporting the hypothesis that mezigdomide-mediated degradation of IKZF1 can reverse T-cell exhaustion not only in vitro but also in patients.

Collectively, our studies provide critical insights into the epigenetic regulation of T-cell exhaustion and demonstrate the potential of mezigdomide in restoring Tex functionality. Bridging preclinical findings to clinical development, our data support ongoing clinical investigations of mezigdomide in combination with BCMA-targeting TCEs such as elranatamab ([ClinicalTrials.gov](#) identifier: NCT06645678).

Authorship

Contribution: H.C., J.Z., and P.R.H. conceptualized and designed the study, acquired, analyzed, and interpreted data, and wrote and edited the manuscript; T.B., C.B., C.-C.H., G.J., M.A., T.A.M., L.C., S.G., S.P.,

A.K.G., and M.O.E. analyzed and interpreted data; and T.-H.H. acquired and analyzed data.

Conflict-of-interest disclosure: H.C., J.Z., C.-C.H., T.B., C.B., T.-H.H., M.A., A.K.G., M.O.E., and P.R.H. are current employees of Bristol Myers Squibb (BMS) and have equity ownership. G.J. is a former employee of the Biocon BMS Research Center. The remaining authors declare no competing financial interests.

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Footnotes

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*H.C. and J.Z. contributed equally to this study.

All epigenetic and transcriptomic data have been deposited in the Gene Expression Omnibus database (accession number GSE301719).

Original data are available from the corresponding author, Patrick R. Hagner (patrick.hagner@bms.com).

The online version of this article contains a data supplement.

There is a [Blood Commentary](#) on this article in this issue.

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